Understanding Lyme Disease
& Co-Infections

A Quick Guide to Tick-Borne Infections

? WHAT IS LYME DISEASE?

Lyme disease is caused by the bacterium Borrelia burgdorferi, transmitted through the bite of infected blacklegged ticks. It's the most common vector-borne illness in the United States, with ~500,000 cases diagnosed annually. Early symptoms include fever, fatigue, and the characteristic 'bull's-eye' rash (erythema migrans).

3 THE "BIG 3" TICK-BORNE PATHOGENS

BORRELIA

(Lyme Disease)

- Spirochete bacterium
- Can affect joints, heart, and nervous system
- Symptoms: fatigue, joint pain, brain fog
- May persist after standard treatment
- Tx: Doxycycline, amoxicillin, ceftriaxone

BABESIA

(Babesiosis)

- Malaria-like parasite
- Infects red blood cells
- Symptoms: sweats, chills, air hunger
- Severe in immuno-compromised patients
- Tx: Atovaquone + azithromycin, or clindamycin + quinine

BARTONELLA

(Bartonellosis)

- Intracellular bacterium
- Also spread by fleas, lice, and cat scratches
- Symptoms: skin striae, neuro/psych issues
- Often underdiagnosed
- Tx: Rifampin + azithromycin or doxycycline

! WHY CO-INFECTIONS MATTER

- Up to 30% of Lyme patients are co-infected with one or more additional pathogens
- Each pathogen requires different treatment — antibiotics don't work on parasites
- Co-infections can make symptoms more severe and harder to diagnose
- Standard Lyme testing does NOT check for Babesia or Bartonella
- Unrecognized co-infections may explain why some patients don't improve with treatment

Sx KEY SYMPTOMS TO RECOGNIZE

LYME	BABESIA	BARTONELLA
Bull's-eye rash	Night sweats	Skin striae/streaks
Joint pain/swelling	Air hunger	Anxiety/irritability
Brain fog	Chills/fever cycles	Burning foot pain
Fatigue	Severe headaches	Swollen lymph nodes

✓ TESTING CONSIDERATIONS

- Standard two-tier Lyme testing can miss up to 50% of early cases
- Consider comprehensive tick-borne disease panels, not just single-pathogen tests
- Specialized labs (Galaxy, IGeneX) offer enhanced testing methods (BAPGM, ddPCR)
- Serology detects antibodies — PCR/DNA testing can detect active infection